

OISD-STD-175

Cementing Operations

Industry: Oil & Natural Gas

Category: Exploration & Production (Upstream)

Edition/Revision: 3rd Edition (Jan 2015) - Originally published 1999, 2nd ed. 2008

Research Date: August 26, 2026

DISCLAIMER: This is an AI-generated research summary based on publicly available information. It is NOT an official OISD publication and must not be considered a replacement for the original OISD standard. The official standard should always be consulted for authoritative requirements.

1. STANDARD OVERVIEW

Standard Number: OISD-STD-175

Official Title: Cementing Operations

Type: Standard (STD)

Industry Category: Exploration & Production / Well Operations

Edition Date: January 2015 (Previous editions: 1999, 2008)

Main Subject: The selection, planning, execution, and equipment safety related to oil well cementing operations. It establishes self-regulatory measures aimed at standardizing the design and execution of cementation jobs across the Indian petroleum industry.

2. PURPOSE OF THE STANDARD

The core objective of OISD-STD-175 is to establish uniform, safe, and reliable practices for cementing oil and gas wells. Cementing is a critical operation because an ineffective cement job can lead to severe structural failures, blowouts, and subsurface environmental contamination.

The standard ensures that the set cement provides lasting structural support for the casing and guarantees absolute zonal isolation (preventing fluids from one geological zone from migrating to another, especially in abnormally pressured gas-bearing formations).

3. SCOPE

Based on publicly available indexes and summaries, the standard broadly covers:

- **Material Requirements:** Oil well cement selection criteria (physical and chemical properties).

- **Operational Phases:** Primary cementing, secondary cementing, and specialized cementing techniques.
- **Special Circumstances:** Cementation of specific well types (e.g., highly deviated, HPHT) and well abandonment procedures.
- **Logistics & Storage:** Handling, storing, and mixing of bulk cement and specialized additives.
- **Equipment Safety:** Standards for cementing units, high-pressure pumps, and associated pipelines.

Exclusion: OISD-175 serves as an industry standard but explicitly notes that it does not supersede the statutory regulations mandated by the Directorate General of Mines Safety (DGMS) or the Chief Controller of Explosives (CCE).

4. INDUSTRY APPLICATION

This standard is highly relevant throughout the upstream lifecycle of well construction and decommissioning:

- **Drilling & Completion:** Primary cementing to secure the casing string after a section is drilled.
- **Well Intervention / Workover:** Secondary (remedial) cementing (like squeeze cementing) to repair casing leaks or isolate depleted zones.
- **Decommissioning:** Placing structural cement plugs for permanent well abandonment.

5. MAJOR AREAS COVERED

- **Cementation Requirements & Planning:** Mandatory issuance of a well-specific cementing plan.
- **Cement & Additive Properties:** Requirements for water quality, cement soundness, free water content, and compressive strength.
- **Cementing Equipment:** Specifications for pumping units, mixing silos, lines, and pressure relief systems.

- **Operational Safety:** Storage protocols, handling of dry cement, and avoiding chemical intermixing.
- **Well Abandonment:** Requirements for securely sealing off wellbores permanently.

6. DETAILED TOPIC-WISE SUMMARY

Cementing Plan and Slurry Properties

What it covers: Pre-job engineering and fluid design.

Important Requirements: A formal "cementing plan" is mandatory for each job. The slurry must be formulated to isolate permeable and high-pressure gas zones effectively. The resulting set cement must offer enduring structural support.

Material Storage and Handling

What it covers: Safe practices for keeping raw cement and chemical additives prior to the operation.

Important Requirements:

- Oldest stock must be issued first (First-In, First-Out).
- Cement chemical bags require elevated storage: at least 6 inches of air space from the ground, kept away from walls, and sealed with a tarpaulin on the base to prevent moisture ingress.
- Air used for bulk transfer must be dry.

Equipment Specifications (Pumping & Mixing)

What it covers: The mechanical reliability of the cementing unit.

Important Requirements: The unit must mix and deliver slurry at highly accurate volumes and densities. A critical safety parameter is that the pumps and associated lines must possess a pressure rating capable of handling **1.5 times the maximum anticipated pumping pressure**. A pressure release safety valve must be installed on the discharge line to prevent catastrophic rupture from sudden pressure spikes.

Well Abandonment Procedure

What it covers: Pumping cement plugs to safely permanently or temporarily abandon a well.

Important Requirements: Ensures that hydrocarbon zones are effectively sealed off so they do not leak to the surface or contaminate freshwater aquifers over decades of abandonment.

7. EQUIPMENT AND SYSTEMS

Equipment / System	Purpose	Standard Requirement (Publicly Known)
Cementing Unit	Mixes dry cement, water, and additives to formulate slurry, then pumps it into the well.	Must deliver precise volume and density; pumps/lines rated to 1.5x max anticipated pressure.
Discharge Lines	Transports high-pressure slurry from the pump to the wellhead.	Must feature a securely placed pressure release safety valve to prevent bursts.
Bulk Storage / Silos	Stores massive quantities of dry cement on site.	Requires dry air for pneumatic transfer to avoid premature hydration or clumping.

8. SAFETY REQUIREMENTS

- **Pressure Safety:** The requirement for equipment to withstand 1.5 times the maximum operational pressure acts as a crucial safety buffer. The mandatory use of a pressure relief valve prevents catastrophic line failures on the rig floor.
- **Chemical Safety:** Emphasizes keeping cement chemicals strictly separate from other rig chemicals to prevent dangerous or detrimental intermixing.

- **Environmental/Well Safety:** Ensuring adequate compressive strength and zonal isolation protects subsurface aquifers from gas and oil contamination.

9. OPERATIONAL REQUIREMENTS

Job Planning: A written cementing plan is an absolute prerequisite. Ad-hoc pumping without engineering calculation is prohibited.

Quality Control: The standard mandates checks on the free water content and compressive strength of the cement. If the cement slurry allows too much free water to separate before setting, it creates a channel for gas migration.

Logistics Controls: Rigorous inventory rotation (oldest stock first) prevents the use of degraded or expired cement which could lead to a failed cementation job.

10. INSPECTION, TESTING & MAINTENANCE

Equipment Checklists: The standard explicitly contains an "Equipment Checklist" section. Although full checklist details are not public, it implies mandatory pre-job verification of cementing head connections, valve integrity, and pump functionality.

Material Testing: Cement selection relies on verifying physical and chemical properties beforehand, including soundness and strength parameters as dictated by the standard.

11. EMERGENCY & HSE ASPECTS

The standard contains specific "Cementing Safety Guide Lines." Ensuring zonal isolation is in itself a primary HSE function—preventing uncontrolled subsurface fluid migration (underground

blowouts). The safe layout of high-pressure lines and relief valves physically protects the crew during the pumping phase, which operates under extreme pressures.

12. IMPORTANT TECHNICAL CONCEPTS

Zonal Isolation: The primary goal of cementing. It means creating a complete hydraulic seal in the annulus between the casing and the geological formation, preventing fluids from moving vertically between different rock layers.

Primary vs. Secondary Cementing:

Primary: The initial operation performed immediately after running a casing string into the well.

Secondary (Remedial): Interventions performed later (like squeeze cementing) to repair flawed primary jobs, seal perforations, or fix casing leaks.

Compressive Strength: The ability of the set cement to withstand the physical crushing forces of the earth and the casing without failing.

13. IMPORTANT TERMS & ABBREVIATIONS

Term / Abbreviation	Meaning	Relevance
Slurry	A pumpable mixture of cement, water, and additives.	The fundamental material of the operation.
OISD	Oil Industry Safety Directorate	The regulatory knowledge center in India issuing this standard.
Free Water	Excess water that separates from the cement slurry before it sets.	Must be strictly minimized to prevent vertical fluid channeling.
DGMS	Directorate General of Mines Safety	

Term / Abbreviation	Meaning	Relevance
		Statutory Indian body whose regulations must be followed alongside OISD.

14. RELATED STANDARDS & REGULATIONS

- **OISD-STD-174:** Well Control Standards (often overlaps during well planning to handle high pressures).
- **OISD-STD-176:** Health, Safety & Environment Training For Exploration & Production Industry (referenced nearby in OISD indices).
- **API Spec 10A / 10B:** Global API standards for cements and materials for well cementing (typically aligned with OISD properties).
- **Petroleum and Natural Gas Rules / DGMS guidelines:** The statutory framework that OISD-175 complements in India.

15. PRACTICAL USE CASES

Scenario 1: Securing a Production Casing

Relevant aspect: Primary Cementing Practices & Slurry Properties.

Why it matters: When a rig completes drilling the pay zone, operators use OISD-175 parameters to design a cement job that ensures the gas reservoir is perfectly isolated from the water aquifers above it.

Scenario 2: Pumping Operations Safety

Relevant aspect: Cementing Equipment (Section 6).

Why it matters: If an unexpected blockage occurs downhole during pumping, pressure will rapidly spike. The OISD-175 requirement for a pressure release safety valve and lines rated for 1.5x expected pressure prevents a fatal surface explosion.

Scenario 3: Site Logistics on an Onshore Rig

Relevant aspect: Material Storage.

Why it matters: During the monsoon season, a rig supervisor enforces the rule of placing cement sacks 6 inches off the ground with a tarpaulin base to ensure the cement doesn't pre-hydrate and ruin the multi-million dollar well.

16. KEY TAKEAWAYS

1. OISD-STD-175 is the authoritative guide for cementing operations in India's oil and gas sector.
2. A detailed cementing plan must be documented for every operation.
3. Zonal isolation and structural support are the ultimate performance metrics for set cement.
4. Safety equipment, including pressure release valves, is mandatory on all discharge lines.
5. Pumping equipment must handle 1.5 times the maximum anticipated pressure.
6. Strict moisture and chemical isolation protocols govern the storage of dry cement and additives.
7. The standard encompasses primary, secondary, specialized, and abandonment cementing.
8. It does not override DGMS or CCE statutory requirements but supplements them.
9. Oldest cement stocks must be consumed first to ensure chemical viability.
10. Continuous updates (1999, 2008, 2015) reflect evolving technologies in well isolation.

17. INFORMATION LIMITATIONS

- **Verified Information:** General scope, historical editions, structural indices, exact safety multipliers for pump pressure (1.5x), storage requirements (6-inch air gap, first-in-first-out), and the necessity of a pre-job cementing plan.
- **Information Requiring Official Access:** The specific numerical limits for "compressive strength," exact formulas for calculating "free water content," specific checklist points, and detailed procedural guides for squeezing operations.

Note: "Detailed requirement is not publicly available in the sources reviewed." No values for required compressive strength or chemical formulas have been fabricated.

18. SOURCES & REFERENCES

1. **Oil Industry Safety Directorate (OISD)** - Official Standards List & Exploration/Production Section. URL: www.oisd.gov.in (Accessed August 26, 2026). [1.3.1, 1.3.2]
2. **Scribd Document Repository** - OISD-STD-175 Cementing Operations (Extracts covering Foreword, Scope, Equipment constraints, Storage protocols, and indices). [1.1.1, 1.1.3]
3. **Mechanical Galaxy / Industry Blogs** - Listings of OISD mechanical and E&P standards. [1.1.2]